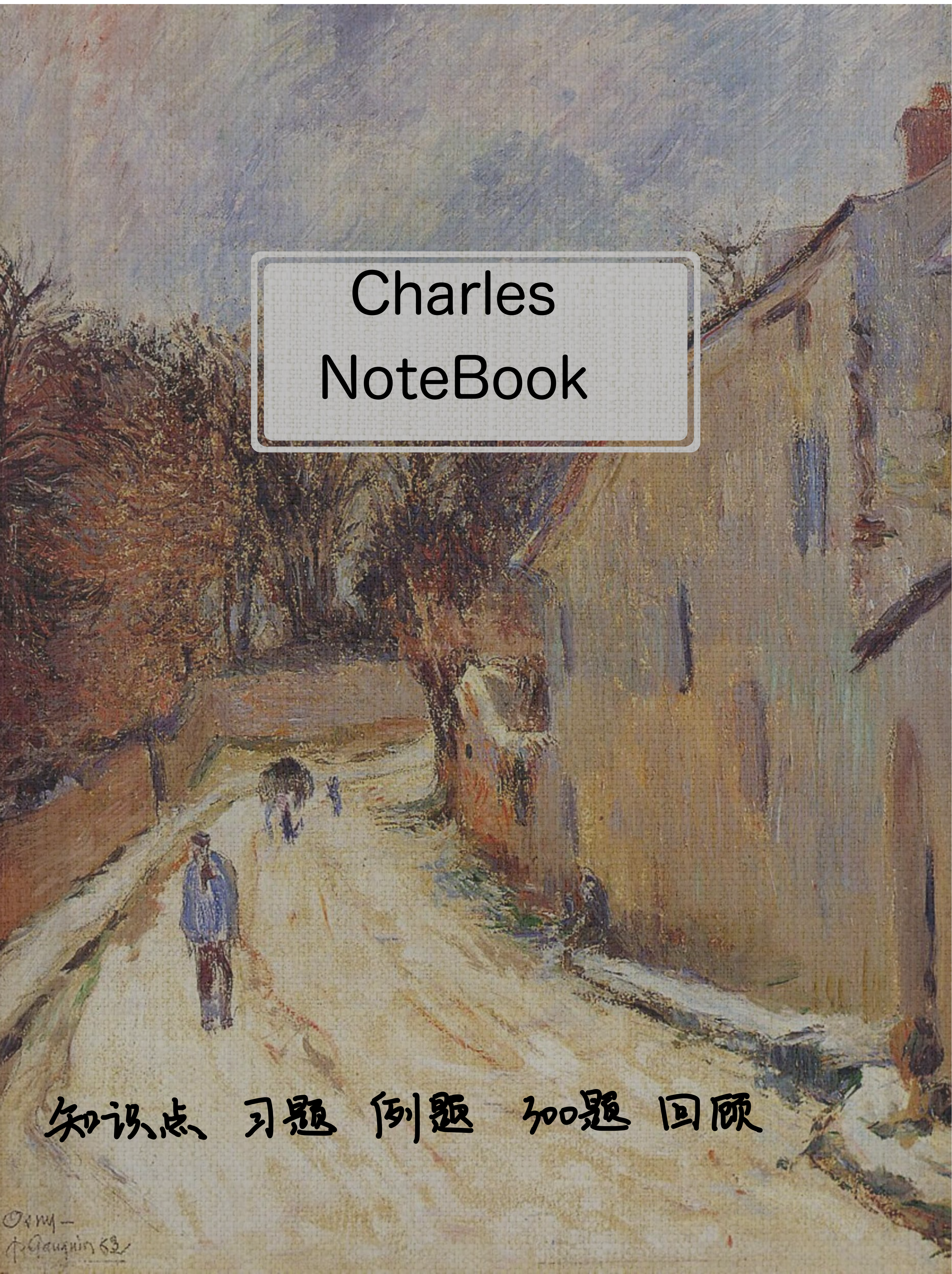


Charles NoteBook

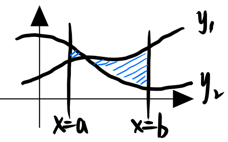
知识点、习题 例题 加题 回顾

面积
体积
平均值

面积

复习第1讲 三大体系下的图形

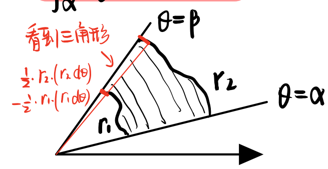
$$S=\int_a^b |y_1(x)-y_2(x)|\,dx$$



- 直角系下直接算 (例9.1)
- 参数方程
  - 直接算(少)
  - 换元法 ★ (例9.2) $\int_\alpha^\beta y(t)\,d\,x(t)$

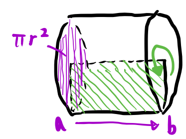
$$S=\frac{1}{2}\int_\alpha^\beta |r_1^2(\theta)-r_2^2(\theta)|\,d\theta$$

极坐标系 例9.3



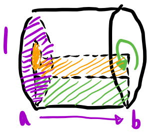
体积

①



$$V=\int_a^b \pi y^2\,dx$$

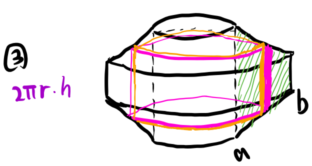
②



$$V=\pi\int_a^b |y_1^2-y_2^2|\,dx$$

例9.8

③



$$V=\int_a^b (2\pi x)\cdot y(x)\,dx$$

柱壳法

平均值

例9.10

$$\bar{y}=\frac{1}{b-a}\int_a^b y(x)\,dx$$

平均高

$$\int_a^b y(x)\,dx = y(\xi)(b-a) \quad a<\xi<b$$

平均值定理

小结:

$S=\int_a^b \square\,dx \qquad S=\frac{1}{2}\int_\alpha^\beta \square^2\,d\theta$

$V_x=\int_a^b \pi\square^2\,dx$

$V_y=\int_a^b 2\pi x\,\square\,dx$

$\bar{y}=\frac{1}{b-a}\int_a^b \square\,dx$

$y=f(x)$

$r=r(\theta)$

如 $y=\sin^4 x$ 在 $[0,\pi/2]$ 上面积: $S=\int_0^{\pi/4}\sin^4 x\,dx=\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$

如 $y=\sqrt{x(1-x)^9}$ 在 $[0,1]$ 上绕X轴体积 $V_x=\int_0^1 \pi y^2\,dx=\int_0^1 \pi x(1-x)^9\,dx \overset{1-x=t}{=} \int_0^1 \pi(1-t)t^9\,dt=\pi\left(\frac{1}{10}-\frac{1}{11}\right)$

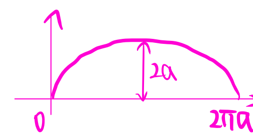
例9.1

求由 $y=\sin x$, $y=\cos x$, $x=0$, $x=\frac{\pi}{2}$ 所围平面图形面积

$$\int_0^{\frac{\pi}{2}} |\cos x-\sin x|\,dx = 2\int_0^{\pi/4} \cos x-\sin x\,dx = 2\left[\sin x+\cos x\right]\Big|_0^{\frac{\pi}{4}} = 2\left(\frac{\sqrt2}{2}+\frac{\sqrt2}{2}-1\right)=2\sqrt2-2$$

例9.2

求由摆线 $\begin{cases} x=a(t-\sin t)\\ y=a(1-\cos t)\end{cases}$ $(a>0)$ 的一拱(见图9-3)与X轴所围平面图形的面积、



$\begin{cases} x=a(t-\sin t)\\ y=a(1-\cos t)\end{cases}$ $(a>0)$

[分析] 参数方程下的题是重点

$$S=\int_0^{2\pi a} f(x)\,dx \quad (\text{直角系下})$$

$$=\int_0^{2\pi} y(t)\,d\,x(t)$$

$\Longrightarrow$ 摆线 $\begin{cases} x=a(t-\sin t)\\ y=a(1-\cos t)\end{cases}$ 一拱面积、$3\pi a^2$

$$=\int_0^{2\pi} a(1-\cos t)\,d\,a(t-\sin t)$$

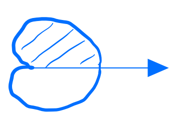
$$=a^2\int_0^{2\pi}(1-\cos t)^2\,dt = a^2\int_0^{2\pi} 1-2\cos t+\cos^2 t\,dt$$

$$=2\pi a^2-2a^2\int_0^{2\pi}\cos t\,dt + a^2\int_0^{2\pi}\cos^2 t\,dt$$

$$=2\pi a^2-0+a^2\cdot\frac12\cdot\frac{\pi}{2}\cdot 4 = 3\pi a^2$$

例9.3

求心形线 $r=a(1+\cos\theta)$ $(a>0)$ 所围平面图形面积、



$$S=2\cdot\frac12\int_0^{\pi} |\overset{a(1+\cos\theta)}{r_2^2}-\overset{0}{r_1^2}|\,d\theta$$

$$=a^2\int_0^{\pi}(1+\cos\theta)^2\,d\theta = a^2\int_0^{\pi} 1+2\cos\theta+\cos^2\theta\,d\theta = a^2\pi+2a^2\cdot\frac12\cdot\frac{\pi}{2} = \frac32 a^2\pi$$

例9.8 ☆☆☆

计算由摆线 $\begin{cases} x=a(t-\sin t)\\ y=a(1-\cos t)\end{cases}$ $(a>0)$ 的一拱与X轴所围平面图形分别绕X轴及y轴旋转一周所得旋转体的体积

绕X轴: $$\int_0^{2a} S\,dx = \pi\int_0^{2a} y^2\,dx = \pi\int_0^{2\pi} a^2(1-\cos t)^2\,d\,a(t-\sin t)$$

$$=a^3\pi\int_0^{2\pi}(1-2\cos t+\cos^2 t)(1-\cos t)\,dt = a^3\pi\int_0^{2\pi}(1-\cos t-2\cos t+2\cos^2 t+\cos^2 t-\cos^3 t)\,dt$$

$$=a^3\pi\int_0^{2\pi}(1-3\cos t+3\cos^2 t-\cos^3 t)\,dt$$

$$=a^3\pi\left[2\pi+3\cdot\frac12\cdot\frac{\pi}{2}\cdot 4\right]=5a^3\pi^2$$

绕y轴 $$\int_0^{2a} 2\pi R\cdot h\,dx = \int_0^{2\pi} 2\pi\cdot a(t-\sin t)\cdot a(1-\cos t)\,d\,a(t-\sin t)$$

$$=2\pi a^3\int_0^{2\pi}(t-\sin t)(1+\overset{1-\sin^2 t}{\cos^2 t}-2\cos t)\,dt$$

$$=2\pi a^3\int_0^{2\pi} t+t\cos^2 t-2t\cos t\underbrace{-2\sin t-\sin^3 t}_{0}+\underbrace{2\sin t\cos t}_{0}\,dt$$

$$=2\pi a^3\left[2\pi^2+\pi^2\right]=6\pi^3a^3$$

例9.10

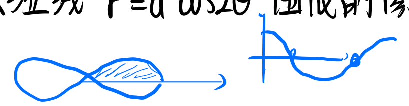
$y=\frac{x}{\sqrt{1-x^2}}$ 在 $[0,\frac{1}{2}]$ 上的平均值

$$\bar{y}=\frac{1}{\frac{1}{2}-0}\int_0^{\frac{1}{2}}\frac{x}{\sqrt{1-x^2}}dx$$

$$=-\int_0^{\frac{1}{2}}\frac{d\,1-x^2}{\sqrt{1-x^2}}=\int_{\frac{3}{4}}^{1}t^{-\frac{1}{2}}dt=2t^{\frac{1}{2}}\Big|_{\frac{3}{4}}^{1}=2-\sqrt{3}$$

例9.4

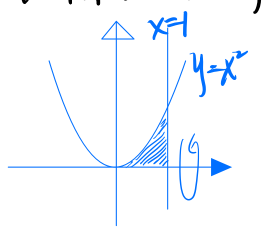
求伯努利双纽线 $r^2=a^2\cos2\theta$ 围成的图形面积



$$S=4\cdot\frac{1}{2}\int_0^{\frac{\pi}{2}}r^2d\theta=2a^2\int_0^{\frac{\pi}{4}}\cos2\theta\,d\theta=2a^2\frac{\sin2\theta}{2}\Big|_0^{\pi/4}=a^2$$

例9.5

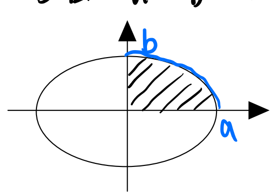
设平面图形由曲线 $y=x^2$ 与直线 $x=1$ 及 x轴围成，求平面图形绕 x 一周 体积



$$\int_0^1\pi(x^2)^2dx=\frac{\pi}{5}$$

例9.6

已知 D 由 $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ 围成，求 D 绕 x 轴旋转一周所得的旋转体体积.



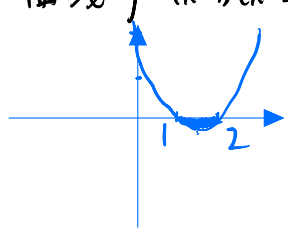
$$y^2=b^2(1-\frac{x^2}{a^2})$$

$$V=2\cdot\int_0^x S\,dx=2\int_0^x\pi y^2dx=2\pi b^2\int_0^a1-\frac{x^2}{a^2}dx$$

$$=2\pi ab^2-\frac{2\pi b^2}{a^2}\frac{a^3}{3}=\frac{4}{3}\pi ab^2$$

例9.7

曲线 $y=(x-1)(x-2)$ 和 x 轴围成一平面图形，求绕 y 轴一周形成的体积



$$V=2\pi\int_1^2-(x-1)(x-2)x\,dx=-2\pi\int_1^2x^3-3x^2+2x\,dx$$

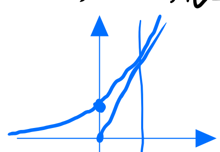
$$=-2\pi\left[\frac{x^4}{4}-x^3+x^2\right]_1^2=-2\pi\left[-\frac{1}{4}\right]=\frac{\pi}{2}$$

例9.9

过(0,0)作曲线 $y=e^x$ 的切线，该切线与曲线 $y=e^x$ 及 x 轴向 x 轴负向伸展的平面记为 D.

求 1) D的面积 A

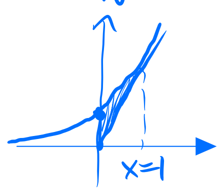
2) D绕直线 $x=1$ 一周的旋转体体积 V



切线 $\begin{cases} y=ax \\ y=e^x \end{cases} \Rightarrow \begin{cases} ① \ y_0=ax_0=e^{x_0} \\ ② \ y_0'=e^{x_0}=a \end{cases} \Rightarrow \begin{cases} x_0=1 \\ y_0=e \\ a=e \end{cases}$

1) $A=\int_0^1 (e^x-ex)\,dx = e^x-\frac{e}{2}x^2\Big|_0^1=\frac{e}{2}$

2)



$$V_1=2\pi\int_0^1(1-x)(e^x-ex)\,dx$$

$$=2\pi\int_0^1 e^x-ex-xe^x+ex^2\,dx$$

$$=2\pi\left[e^x-\frac{e}{2}x^2-(x-1)e^x+\frac{e}{3}x^3\right]_0^1$$

$$=2\pi\left[e-\frac{e}{2}+\frac{e}{3}-1-1\right]=2\pi\left[\frac{5}{6}e-2\right]=\frac{5}{3}\pi e-4\pi$$

$$V_2=2\pi\int_{-\infty}^0(1-x)e^x\,dx$$

$$=2\pi\left[\int_{-\infty}^0 e^x dx-\int_{-\infty}^0 xe^x dx\right]$$

$$=2\pi\left[(2-x)e^x\right]_{-\infty}^0=2\pi\left[2-\lim_{x\to-\infty}xe^x\right]=4\pi$$

$$\Rightarrow V=V_1+V_2=\frac{5}{3}\pi e$$

习9.1

已知曲线 $y=a\sqrt{x}\ (a>0)$ 与曲线 $y=\ln\sqrt{x}$ 在点 (x_0,y_0) 处有公共切线，求：

(1) 常数 a 及切点 (x_0,y_0)

(2) 两曲线与 x 轴围成的平面图形的面积 S

(1) $y_1=a\sqrt{x}$

$y_1'=\frac{a}{2}x^{-\frac{1}{2}}$

$y_2=\ln\sqrt{x}$

$y_2'=\frac{1}{\sqrt{x}}\cdot\frac{1}{2}x^{-\frac{1}{2}}=\frac{1}{2x}$

$\begin{cases} a\sqrt{x_0}=\ln\sqrt{x_0} \\ \frac{a}{2\sqrt{x_0}}=\frac{1}{2x_0} \end{cases} \Rightarrow a=\frac{1}{\sqrt{x_0}}$

$\ln\sqrt{x_0}=1 \Rightarrow x_0=e^2$

$y_0=1$

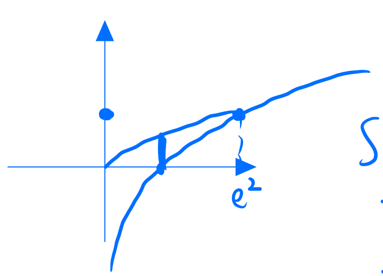
$a=\frac{1}{e}$

$(x_0,y_0)=(e^2,1)$

$y_1=\frac{1}{e}\sqrt{x}$

$y_2=\ln\sqrt{x}$

(2).



$$S=S_1+S_2=\int_0^1\frac{\sqrt{x}}{e}dx+\int_1^{e^2}\frac{\sqrt{x}}{e}-\frac{1}{2}\ln x\,dx$$

$$=\frac{1}{e}\int_0^{e^2}\sqrt{x}\,dx-\frac{1}{2}\int_1^{e^2}\ln x\,dx$$

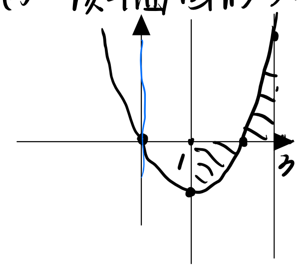
$$=\frac{2}{3}e^2-\frac{1}{2}\left[x\ln x\Big|_1^{e^2}-\int_1^{e^2}1\,dx\right]=\frac{1}{6}e^2-\frac{1}{2}$$

习9.2

设曲线 $y=x^2-2x$ ($1\le x\le 3$), $y=0$, $x=1$, $x=3$ 围成平面图形A,求

(1) A的面积S

(2) 该平面图形绕y V.



(1) $S=\int_1^2 2x-x^2\,dx+\int_2^3 x^2-2x\,dx$

$= x^2-\frac{x^3}{3}\Big|_1^2 + \frac{x^3}{3}-x^2\Big|_2^3$

$=(4-\frac{8}{3})-(1-\frac{1}{3})+(\frac{27}{3}-9)-(\frac{8}{3}-4)$

$=8-\frac{16}{3}-\frac{2}{3}=2$

(2) $V=2\pi\left[\int_1^2 x(2x-x^2)\,dx+\int_2^3 x(x^2-2x)\,dx\right]$

$=2\pi\left[\int_1^2 2x^2-x^3\,dx+\int_2^3 x^3-2x^2\,dx\right]$

$=2\pi\left[\frac{2}{3}x^3-\frac{x^4}{4}\Big|_1^2+\frac{x^4}{4}-\frac{2}{3}x^3\Big|_2^3\right]$

$=2\pi\left[(\frac{16}{3}-4)-(\frac{2}{3}-\frac{1}{4})+(\frac{81}{4}-18)-(4-\frac{16}{3})\right]$

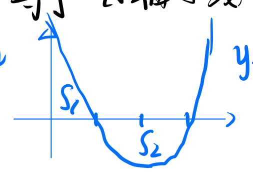
$=2\pi\left[\frac{32}{3}-8-\frac{5}{12}+\frac{81}{4}-18\right]=\pi\cdot\frac{128-26\times12-5+81\times3}{6}=9\pi$

习9.3

已知抛物线过 A(1,0) B(3,0)

(1) 证明两坐标轴与该抛物线所围面积等于X轴与线所围S.

(1)



$y=a(x-1)(x-3)=a(x^2-4x+3)$

$S_1=\int_0^1 a(x^2-4x+3)\,dx$

$=a\left[\frac{x^3}{3}-2x^2+3x\right]_0^1$

$=a\left[\frac{1}{3}-2+3\right]$

$=\frac{4}{3}a$

$S_2=\int_1^3 a(-x^2+4x-3)\,dx$

$=a\left[-\frac{x^3}{3}+2x^2-3x\right]_1^3$

$=a\left[-9+18-9+\frac{1}{3}-2+3\right]$

$=\frac{4}{3}a$

(2) 计算1中两个图形绕x轴旋转一周所得体积之比

$V_1=\pi\int_0^1 a^2(x-1)^2(x-3)^2\,dx$

$=\pi a^2\int_0^1 (x-1)^2(x-3)^2\,dx$

$=\pi a^2\int_0^1 x^4-8x^3+22x^2-24x+9\,dx$

$=\pi a^2\left[\frac{x^5}{5}-2x^4+\frac{22}{3}x^3-12x^2+9x\right]_0^1$

$=\pi a^2\left[\frac{1}{5}-2+\frac{22}{3}-12+9\right]$

$=\pi a^2\frac{38}{15}$

$V_2=\pi\int_1^3 a^2(x-1)^2(x-3)^2\,dx$

$=\pi a^2\int_1^3 (x-1)^2(x-3)^2\,dx$

$=\pi a^2\int_1^3 x^4-8x^3+22x^2-24x+9\,dx$

$=\pi a^2\left[\frac{x^5}{5}-2x^4+\frac{22}{3}x^3-12x^2+9x\right]_1^3$

$=\pi a^2\left[\frac{243}{5}-162+90+27-\frac{38}{15}\right]$

$=\pi a^2\frac{16}{15}$

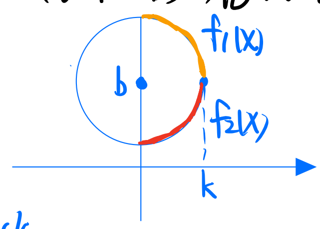
$\Rightarrow$ 比 $\frac{19}{8}$

习9.4

求圆域 $x^2+(y-b)^2\leqslant k^2$ $(0<k<b)$ 绕 x 轴 体积

f_1: $y_1 = b+\sqrt{k^2-x^2}$

f_2: $y_2 = b-\sqrt{k^2-x^2}$



$$V = 2\pi\int_0^k y_1^2 - y_2^2\,dx = 2\pi\int_0^k (b^2+k^2-x^2+2b\sqrt{k^2-x^2}) - (b^2+k^2-x^2-2b\sqrt{k^2-x^2})\,dx$$

$$= 2\pi\int_0^k 4b\sqrt{k^2-x^2}\,dx = 8\pi b\int_0^k \sqrt{k^2-x^2}\,dx \xlongequal{x=k\sin t} 8\pi bk\int_0^{\frac{\pi}{2}} \cos t\, d\,k\sin t$$

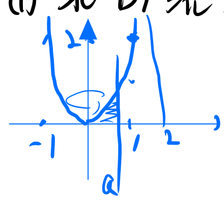
$$= 8\pi bk^2\int_0^{\frac{\pi}{2}}\cos^2 t\,dt = 8\pi bk^2\,\frac{1}{2}\cdot\frac{\pi}{2} = 2\pi^2k^2b$$

习9.5

设 D_1 是由 $y=2x^2$ 与 $x=a$, $x=2$, $y=0$ 所围成平面区域

D_2 是由 $y=2x^2$ 与 $y=0$, $x=a$ 所围平面区域, $0<a<2$

(1) 求 D_1 绕 x 轴 V_1 D_2 绕 y 轴 V_2



$$V_1 = \pi\int_a^2 (2x^2)^2\,dx = \pi\frac{4}{5}x^5\Big|_a^2 = \frac{4\pi}{5}(32-a^5)$$

$$V_2 = 2\pi\int_0^a x\cdot 2x^2\,dx = 4\pi\frac{x^4}{4}\Big|_0^a = \pi a^4$$

(2) $a=?$ V_1+V_2 max

$$V = V_1+V_2 = \pi a^4 - \frac{4\pi}{5}a^5 + \frac{4\pi}{5}\cdot 32$$

$$V' = 4\pi a^3 - 4\pi a^4 = 4\pi a^3(1-a) \xlongequal{令} 0 \qquad a=0,\ a=1$$

a	0	(0,1)	1	(1,2)	2
V'	0	+	0	−	−
V		↗		↘	

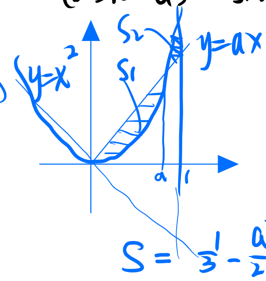
$$V(1) = \frac{\pi}{5} + \frac{4\pi}{5}\cdot 32 \qquad , a=1$$

$$= \frac{129}{5}\pi$$

习9.6

直线 $y=ax$ 与抛物线 $y=x^2$ 所围 S_1,
它们与直线 $x=1$ 所围 S_2. $a<1$

(1) 求 a, s.t. S_1+S_2 min. min?

① 

$ax_0=x_0^2 \Rightarrow x_0=a \Rightarrow y_0=a^2$

$S_1=\int_0^a (ax-x^2)\,dx = \frac{a}{2}x^2-\frac{x^3}{3}\Big|_0^a = \frac{a^3}{6}$

$S_2=\int_a^1 (x^2-ax)\,dx = \frac{x^3}{3}-\frac{a}{2}x^2\Big|_a^1 = \frac{1}{3}-\frac{a}{2}+\frac{a^3}{6}$

$S=\frac{1}{3}-\frac{a}{2}+\frac{a^3}{3}$

$S'=a^2-\frac{1}{2}\overset{令}{=}0 \Rightarrow a=\frac{\sqrt{2}}{2}$

$S_{min}=S(\frac{\sqrt{2}}{2})=\frac{1}{3}-(\frac{\sqrt{2}}{4}-\frac{\sqrt{2}}{12})=\frac{2-\sqrt{2}}{6}$

② $a\le 0$时

$S_1=\int_a^0 (ax-x^2)\,dx = \frac{ax^2}{2}-\frac{x^3}{3}\Big|_a^0 = -(\frac{a^3}{2}-\frac{a^3}{3}) = -\frac{a^3}{6}$

$S_2=\int_0^1 (x^2-ax)\,dx = \frac{x^3}{3}-\frac{ax^2}{2}\Big|_0^1 = \frac{1}{3}-\frac{a}{2}$

$S=\frac{1}{3}-\frac{a}{2}-\frac{a^3}{6}$ $\quad S'=-\frac{1}{2}-\frac{a^2}{2}<0$

$S(0)=S_{min}=\frac{1}{3}$

$\Rightarrow S_{min}=\frac{2-\sqrt{2}}{6}$

(2) 求该最小值所对应的平面图形绕 x 轴旋转一周所得旋转体体积

$V=\pi\left[\int_0^{\frac{\sqrt{2}}{2}} \frac{x^2}{2}-x^4\,dx + \int_{\frac{\sqrt{2}}{2}}^1 x^4-\frac{x^2}{2}\,dx\right]$

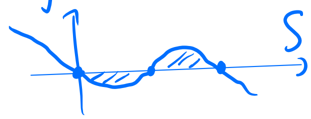
$=\pi\left[\frac{x^3}{6}-\frac{x^5}{5}\Big|_0^{\frac{\sqrt{2}}{2}} + \frac{x^5}{5}-\frac{x^3}{6}\Big|_{\frac{\sqrt{2}}{2}}^1\right]$

$=\pi\left[(\frac{\sqrt{2}}{24}-\frac{\sqrt{2}}{40})2+\frac{1}{5}-\frac{1}{6}\right]=\pi\left[\frac{\sqrt{2}}{30}+\frac{1}{30}\right]=\frac{\sqrt{2}+1}{30}\pi$

300.9.1

$y=x(x-1)(2-x)$ 与x轴所围面积

$y=-x(x-1)(x-2)$



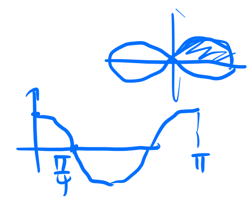
$$S=\int_0^1 x(x-1)(x-2)\,dx-\int_1^2 x(x-1)(x-2)\,dx$$
$$=-\int_0^1 x(x-1)(2-x)\,dx+\int_1^2 x(x-1)(2-x)\,dx$$

(D)

300.9.2

双纽线 $(x^2+y^2)^2=x^2-y^2$ 所围成的面积

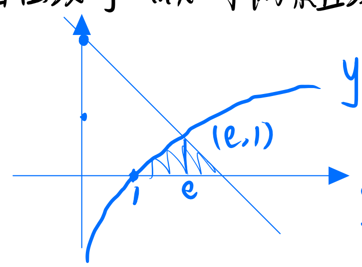
$$r^4(\cos^2\theta+\sin^2\theta)^2=r^2(\cos^2\theta-\sin^2\theta)=(\cos^2\theta-(1-\cos^2\theta))r^2$$
$$r^2=2\cos^2\theta-1=\cos2\theta=r^2 \qquad r=\sqrt{\cos2\theta}$$



~~A~~ (B) $\quad \cancel{4\int_0^{\frac{\pi}{4}}\cos2\theta\,d\theta} \quad 2\int_0^{\frac{\pi}{4}}\cos2\theta\,d\theta$ ☆ 极坐标系有 $\frac{1}{2}$!

300.9.3

由曲线 $y=\ln x$ 与两条直线 $y=e+1-x$ 及 $y=0$ 所围成的平面图形的面积

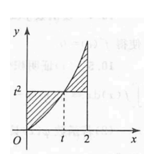


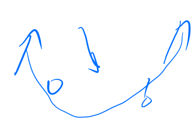
$e+1-x=\ln x$
$\Rightarrow x=e$

$$S=\int_1^e \ln x\,dx+\int_e^{e+1}(e+1-x)\,dx$$
$$=x\ln x\Big|_1^e-(e-1)+e+1-\frac{x^2}{2}\Big|_e^{e+1}$$
$$=1+e-\frac{1}{2}(e^2+1+2e-e^2)$$
$$=\frac{3}{2}$$

300.9.4

设 $y=x^2$ 定义在 $[0,2]$ 上，t为 $[0,2]$ 上任意一点，如图所示，问当t取何值时，图中阴影部分面积最小？





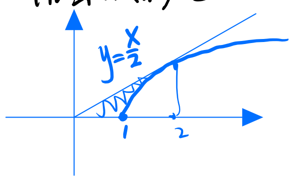
$$S=\int_0^t(t^2-x^2)\,dx+\int_t^2(x^2-t^2)\,dx$$
$$=t^2x-\frac{x^3}{3}\Big|_0^t+\frac{x^3}{3}-t^2x\Big|_t^2$$
$$=t^3-\frac{t^3}{3}+\frac{8}{3}-2t^2-\frac{t^3}{3}+t^3$$
$$=\frac{4}{3}t^3+\frac{8}{3}-2t^2$$

$S_1=\frac{4}{3}+\frac{8}{3}-2=2$

$S'=4t^2-4t=4t(t-1)$ $\qquad S''=8t-4\Rightarrow S_1''=4>0$ 极小

300. 9.5

设曲线 $y=\sqrt{x-1}$，过原点作切线，求此切线与该曲线及x轴所围图形 S

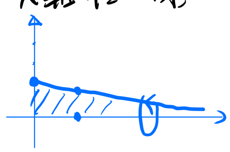


$$y=(x-1)^{\frac{1}{2}}$$
$$y'=\frac{1}{2}(x-1)^{-\frac{1}{2}}$$
$$y_0=\frac{1}{2}(x_0-1)^{-\frac{1}{2}}x_0=(x_0-1)^{\frac{1}{2}}$$
$$\frac{x_0}{2}=x_0-1 \Rightarrow x_0=2,\ y_0=1$$

$$S=\int_0^1 \frac{x}{2}dx+\int_1^2 \frac{x}{2}-\sqrt{x-1}\,dx$$
$$=\frac{x^2}{4}\Big|_0^2-\int_1^2\sqrt{x-1}\,dx$$
$$=1-\frac{2}{3}(x-1)^{\frac{3}{2}}\Big|_1^2=1-\frac{2}{3}=\frac{1}{3}$$

300. 9.6

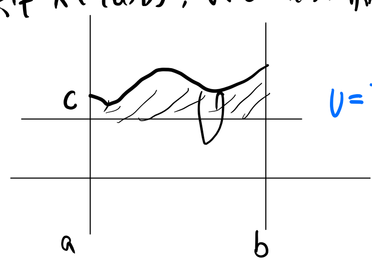
位于曲线 $y=\frac{1}{\sqrt{1+x^2}}$ $(0\le x<+\infty)$ 下方，x轴上方的无界区域绕x轴转一周 V?



$$V=\pi\int_0^{+\infty}y^2\,dx$$
$$=\pi\int_0^{+\infty}\frac{1}{1+x^2}dx=\pi\arctan x\Big|_0^{+\infty}=\pi\left(\frac{\pi}{2}-0\right)=\frac{\pi^2}{2}$$

300. 9.7

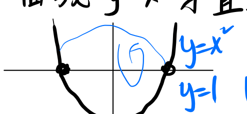
连续曲线段 $y=f(x)$，直线 $x=a$，$x=b$ 及 $y=c$ 围成封闭图形D，其中 $x\in(a,b)$，$0<c<f(x)$，则D绕 $y=c$ 旋转一周 $V=?$



$$V=\pi\int_a^b\left(f(x)-c\right)^2dx$$ (B)

300. 9.8

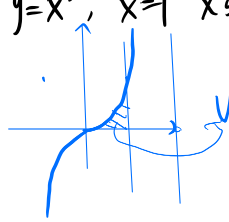
曲线 $y=x^2$ 与直线 $y=1$ 围成的封闭图形绕 $y=1$ V?



$$V=2\pi\int_0^1(1-x^2)^2dx=2\pi\int_0^1 1+x^4-2x^2\,dx=2\pi+2\pi\left[\frac{x^5}{5}-\frac{2}{3}x^3\right]_0^1$$
$$=2\pi+2\pi\left[\frac{1}{5}-\frac{2}{3}\right]$$
$$=\frac{16}{15}\pi$$

300. 9.9

$y=x^3$，$x=1$ x轴 → $x=2$ V?

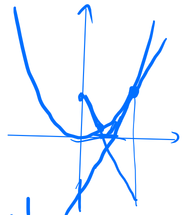


$$V=\int_0^1 2\pi(2-x)x^3dx=\pi\int_0^1 4x^3dx-2\pi\int_0^1 x^4dx=\pi\cdot x^4\Big|_0^1-2\pi\frac{x^5}{5}\Big|_0^1$$
$$=\pi-\frac{2\pi}{5}=\frac{3}{5}\pi$$

300.9.10

过 $y=x^2$ 上一点 $(1,1)$ 作切线，求切线方程，并求该切线与

与曲线 及x轴 绕X轴 V



$y'=2x \quad y'_1=2$

切线：$y=2x-1$

$\begin{cases} y=x^2 \\ y=-2x+1 \end{cases} \Rightarrow x^2+2x-1=0$

$x_0=\dfrac{-2+\sqrt{4+4}}{2}=\sqrt{2}-1$

$$V=\pi\int_0^{\sqrt2-1}(2x-1)^2dx+\pi\int_{\sqrt2-1}^{1}(x^2)^2dx$$

$$=\pi\left[\int_0^{\sqrt2-1}4x^2-4x+1\,dx+\int_{\sqrt2-1}^1 x^4dx\right]$$

$$=\pi\left[\frac{4}{3}x^3-2x^2+x\Big|_0^{\sqrt2-1}+\frac{1}{5}x^5\Big|_{\sqrt2-1}^1\right]$$

$$=\pi\left[\frac43(\sqrt2-1)^3-2(\sqrt2-1)^2+(\sqrt2-1)+\frac15-\frac15(\sqrt2-1)^5\right]$$

$$=\pi\left[\frac43(5\sqrt2-7)-2(3-2\sqrt2)+(\sqrt2-1)+\frac15-\frac15(29\sqrt2-41)\right]$$

$$=\frac{\pi}{15}\left[100\sqrt2-140-90+60\sqrt2+15\sqrt2-15+3-87\sqrt2+123\right]$$

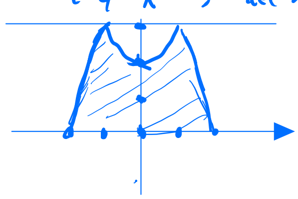
$$=\frac{\pi}{15}\left[88\sqrt2-119\right]$$

$$V=\pi\int_0^{\frac12}x^4dx+\pi\int_{\frac12}^1 x^4-(2x-1)^2dx=\frac{1}{30}\pi$$

300.9.11

求 $y=3-|x^2-1|$ 与x轴所围图形 绕 $y=3$ V

$$y=\begin{cases}2+x^2, & x\in(-1,1)\\ 4-x^2, & x\in(-\infty,-1)或(1,+\infty)\end{cases}$$



$$V=2\pi\left[\int_0^1 9-(2+x^2-3)^2dx+\int_1^2(4-x^2-3)^2dx\right]$$

$$=2\pi\left[\int_0^1 8-x^4+2x^2dx+\int_1^2 1+x^4-2x^2dx\right]$$

$$=2\pi\left[8x-\frac{x^5}{5}+\frac23x^3\Big|_0^1+x+\frac{x^5}{5}-\frac23x^3\Big|_1^2\right]$$

$$=2\pi\left[8-\frac15+\frac23+\left(2+\frac{32}{5}-\frac{16}{3}\right)-\left(1+\frac15-\frac23\right)\right]$$

$$=2\pi\left[9-\frac25+\frac43+\frac{32}{5}-\frac{16}{3}\right]=12\pi$$